How Much of Summit Success Is Decided Before the Climb?

Predicting individual summit success on Himalayan expeditions

Dávid Zoltán Dócs Dávid Szaniszló

Statistical and Machine Learning
17 November 2026



Photo: Mount Everest, north face (L. Galuzzi, CC BY-SA 2.5)

Motivation and research question

Context

- A Himalayan expedition takes weeks and costs tens of thousands of euros
- About one in three climbers reaches the summit
- Peak, season, team and hired support are fixed months in advance

Dataset

- Near-complete record of expeditions in Nepal, 55,049 climbers
- Non-linear effects and interactions
- Two methodological issues: leakage and clustered observations

Research question

Using only information available before departure, how well can we predict whether a climber reaches the summit, and which factors drive the probability?

Binary classification: $Y_i \in \{0, 1\}$, estimate $p(x) = P(Y = 1 \mid X = x)$.

1. How well can we predict?
2. Which method performs best?
3. Which factors matter?
4. How reliable are the predictions?

Data

- Expedition archive of the Nepal Himalaya compiled by Elizabeth Hawley, 1905–2019
- Three tables, members (76,519), expeditions (10,364) and peaks (468), joined to one row per climber
- Target: `success` of the individual climber
- Hired staff are excluded: they reach the summit 65% of the time, as part of their job
- Sample: 55,049 climbers, 9,883 expeditions, 381 peaks; summit rate 33.8%

Filter step	Climbers
All member records	76,519
Year 1950 or later	76,099
Not hired staff	60,533
Climbing role (no doctors, film crews)	56,443
Expedition attempted the climb	55,062
Peak, sex and season known	**55,049**

Table 1: Construction of the analysis sample.

Exploratory analysis

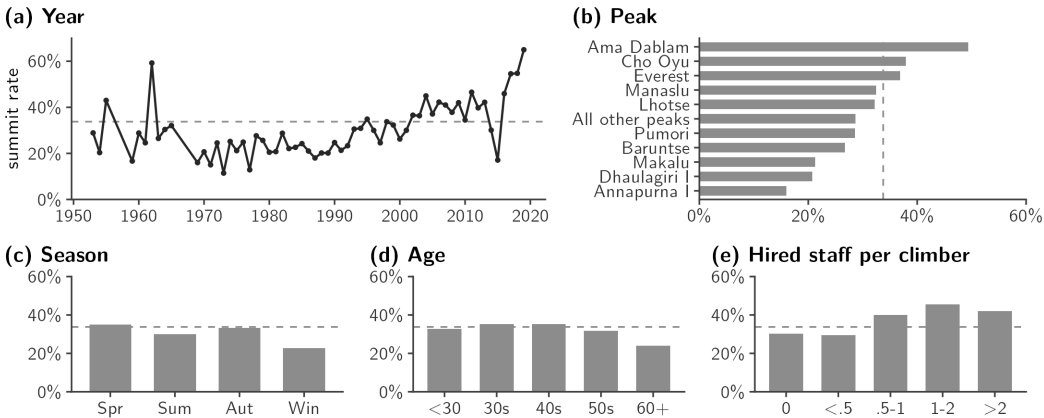


Figure 1: Summit rate by year, peak, season, age and hired staff per climber. Dashed: overall rate (34%).

Pre-processing

Predictors (16)

- Mountain: peak (top 10 + other), height, 8000 m peak, climbed before
- Time: year, season
- Team and support: hired staff, staff per climber, commercial agency, team size, solo
- Climber: age (median imputation + missing indicator), sex, citizenship (top 15 + other), leader

Excluded as leakage

- highest point reached, death, injury, termination reason
- oxygen use: recorded high on the mountain, so it reveals how far the climber got (Figure 2)

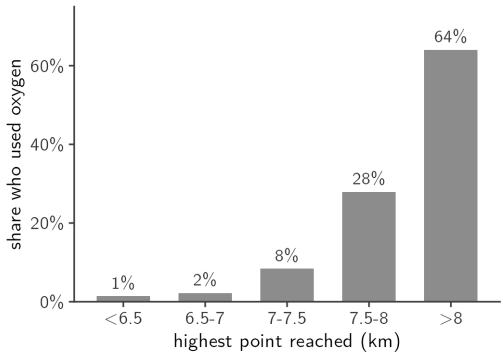


Figure 2: Oxygen use among climbers who failed on 8000 m peaks, by highest point reached.

Validation design and evaluation metrics

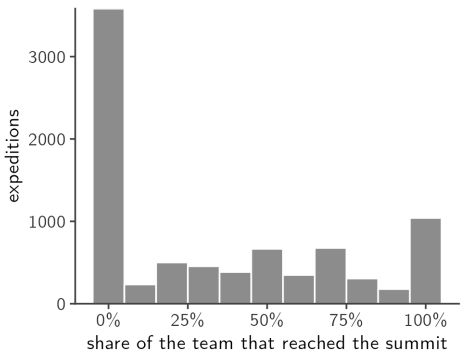


Figure 3: Share of the team reaching the summit, per expedition.

Sample splitting

- Outcomes are clustered within expeditions (Figure 3): in 43% nobody summits, in 12% everybody does
- Test set: 20% of *expeditions* (11,133 climbers), used once
- Tuning: 5-fold CV with folds formed by expedition
- Imputation and category grouping estimated on the training set

Metrics

- Log score (main) and Brier score, both strictly proper:

$$\text{LS} = -\frac{1}{N}\sum_i \left[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \right]$$
$$\text{BS} = \frac{1}{N}\sum_i (\hat{p}_i - y_i)^2$$

- AUC for discrimination, calibration plots
- Accuracy is of limited use: predicting "no summit" for everyone gives 68%

Methods

Method	Model / objective	Tuning (grouped 5-fold CV)	Selected
Elastic-net logistic regression	$\min_\beta -\frac{1}{N}\ell(\beta) + \lambda\left[\frac{1-\alpha}{2}\|\beta\|_2^2 + \alpha\|\beta\|_1\right]$; natural splines for year, age and height	$\alpha \in \{0, .25, .5, .75, 1\}$, λ path, 1-SE rule	$\alpha = 0.25$, 38 of 49 terms
Classification tree	Cost-complexity pruning, $C_{cp}(T) = R(T) + cp\,\|T\|$	cp, 1-SE rule	30 leaves
Random forest	Average of $B = 500$ trees grown on bootstrap samples, m of 16 predictors tried at each split	$m \in \{2, 4, 7, 11\}$ × minimum node size $\in \{25, 100, 400, 1000\}$	$m = 7$, node size 100
Gradient boosting	$f_M(x) = f_{M-1}(x) + \nu\, T(x; \Theta_M)$, Bernoulli deviance	depth J× shrinkage ν× trees M	$J = 6$, $\nu = 0.02$, $M = 2{,}850$

Table 2: Methods, tuning grids and selected values (training set only).

Tuning

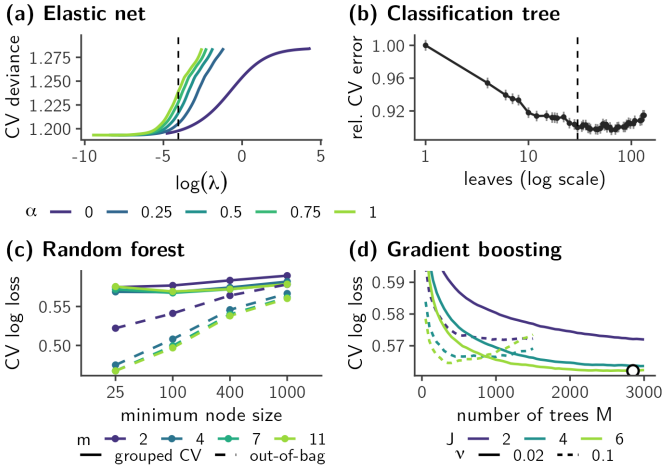


Figure 4: Grouped 5-fold CV error. Vertical lines: 1-SE choice; circle: selected setting.

- Elastic net: α hardly matters; 38 of 49 terms kept
- Tree: 30 leaves
- Forest: out-of-bag error (dashed) too optimistic, 0.50 vs. 0.57, since team-mates stay in the bootstrap sample
- Boosting: deepest trees ($J = 6$) win, so interactions matter

Model assessment on the test set

Model	Log loss	Brier	AUC	Accuracy
Summit rate only	0.629	0.219	0.500	0.678
Elastic net	0.582	0.198	0.695	0.697
Pruned tree	0.579	0.196	0.685	0.716
Random forest	0.543	0.181	0.748	0.736
Boosting	**0.533**	**0.178**	**0.757**	**0.738**

Table 3: Test-set performance (11,133 climbers).

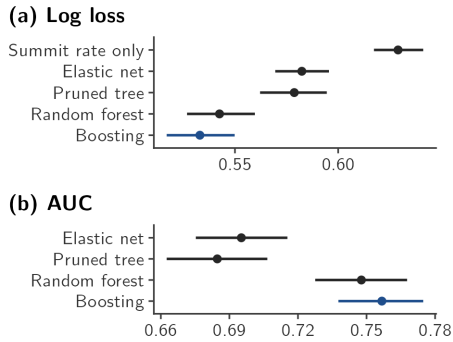


Figure 5: Test metrics with 95% bootstrap intervals.

- Random forest and boosting clearly outperform the elastic net and the single tree
- Boosting vs. random forest: difference in log loss 0.010, 95% CI $[0.003, 0.016]$

ROC curves and confusion matrix

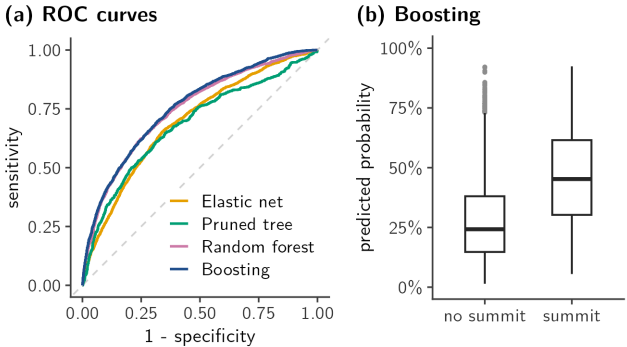


Figure 6: (a) ROC curves. (b) Boosting predictions by observed outcome.

	Observed	
Predicted	Summit	No summit
Summit	1,531	863
No summit	2,050	6,689

Table 4: Boosting confusion matrix, cut-off 0.5.

- Sensitivity 0.43, specificity 0.89
- Precision 0.64
- Cut-off at the summit rate (0.34): sensitivity and specificity both 0.69

Predicted and observed summit rates

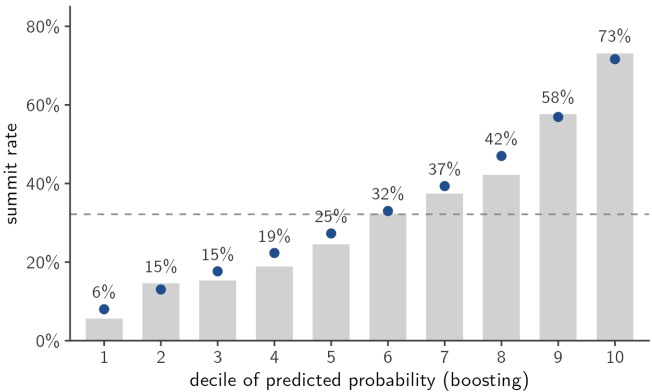


Figure 7: Observed (bars) and predicted (dots) summit rate by decile of the boosting prediction.

- Without a model, every climber gets 34%
- The model separates test climbers into groups with 6% to 73% observed success
- Predicted and observed rates agree in every decile, so the model is well calibrated
- Log loss is 15% below the baseline; much of the outcome remains unexplained

Interpretable models: logit coefficients and pruned tree

Term	$\hat{\beta}$	Odds ratio
log(1 + staff per climber)	0.58	1.79
Peak: Cho Oyu	0.40	1.50
Peak: Annapurna I	−0.36	0.70
Peak: Everest	−0.26	0.77
Season: winter	−0.16	0.85
Role: leader	0.15	1.16
Citizenship: China	0.56	1.75
Citizenship: South Korea	−0.42	0.66

Table 5: Elastic net at $\lambda_{1\text{-SE}}$. Reference: autumn, other peaks, climber, other citizenship.

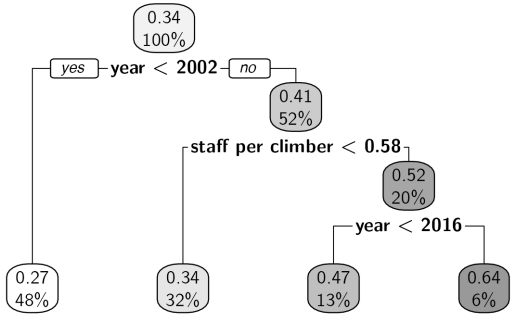


Figure 8: Pruned tree, first splits. Nodes: summit rate and share of climbers.

Boosting: importance and partial dependence

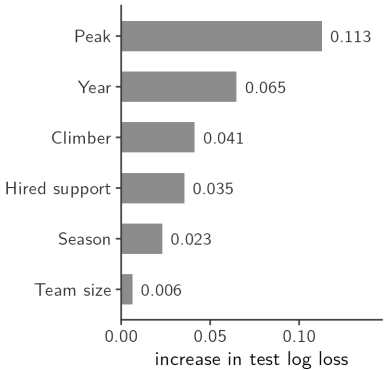


Figure 9: Permutation importance by group of predictors.

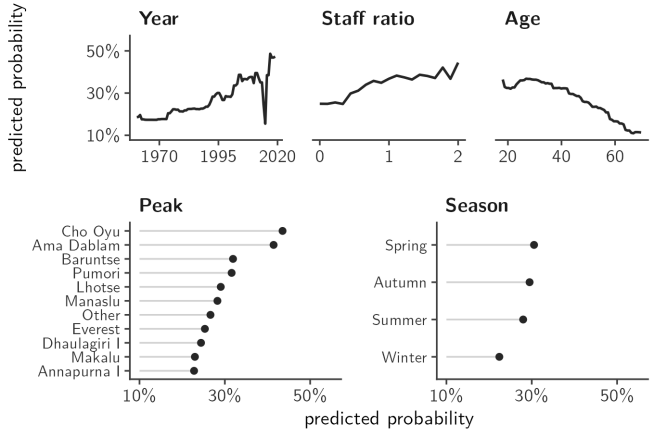


Figure 10: Partial dependence of the boosting prediction (dip in 2015: earthquake).

Robustness checks

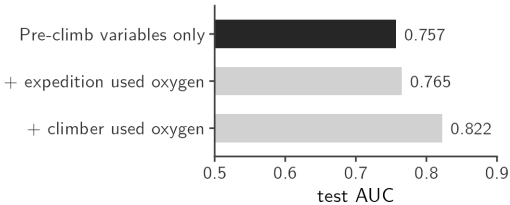


Figure 11: Boosting test AUC with oxygen variables added.

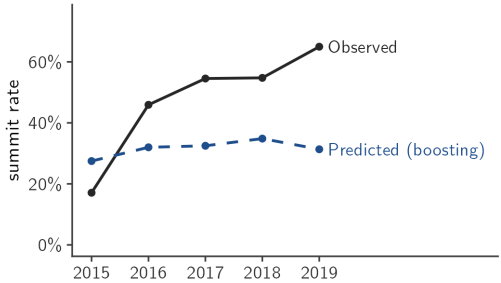


Figure 12: Trained on 1950–2014, predicted for 2015–2019.

- Adding oxygen use raises the AUC from 0.76 to 0.82
- The gain comes from information recorded during the climb, so it is not available before departure

- The ranking still works (AUC 0.63–0.66)
- The level is too low (32% vs. 47%)
- Trees cannot extrapolate the trend, so the model needs refitting for new seasons

Conclusion and limitations

Findings

1. Pre-climb information separates climbers into groups with 6% to 73% success; boosting reaches an AUC of 0.76 and is well calibrated
2. Boosting $>$ random forest $>$ tree $\approx$ elastic net; the gain of the flexible models indicates non-linear effects and interactions
3. Peak and year matter most, followed by age, citizenship and hired support
4. Grouped validation and the removal of leaky variables change the results (OOB 0.50 vs. CV 0.57; leaky AUC 0.82)

Limitations

- No data on weather, health or experience of the climber
- Strong changes over time limit forecasts for new seasons
- Associations only, no causal effects
- Nepal only, data up to spring 2019

References

Breiman, L. (2001). Random forests. *Machine Learning* 45, 5–32.

Friedman, J. (2001). Greedy function approximation: a gradient boosting machine. *Annals of Statistics* 29(5), 1189–1232.

Friedman, J., Hastie, T. and Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software* 33(1).

Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning*, 2nd ed. Springer.

Hawley, E. and Salisbury, R. *The Himalayan Database*. himalayandatabase.com

James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021). *An Introduction to Statistical Learning*, 2nd ed. Springer.

Mock, T. (2020). TidyTuesday 2020-09-22: Himalayan climbing expeditions. github.com/rfordatascience/tidytuesday

Ridgeway, G. and GBM Developers. `gbm`: Generalized boosted regression models. R package.

Therneau, T. and Atkinson, B. `rpart`: Recursive partitioning and regression trees. R package.

Wright, M. N. and Ziegler, A. (2017). `ranger`: A fast implementation of random forests. *Journal of Statistical Software* 77(1).

Title photo: Galuzzi, L. (2006). *Everest North Face toward Base Camp, Tibet*. Wikimedia Commons, CC BY-SA 2.5.

Thank you!

Backup

Backup: test results with 95% bootstrap intervals

Model	Log loss	AUC
Summit rate only	0.629 [0.617, 0.641]	0.500
Elastic net	0.582 [0.570, 0.596]	0.695 [0.675, 0.715]
Pruned tree	0.579 [0.562, 0.594]	0.685 [0.663, 0.707]
Random forest	0.543 [0.527, 0.560]	0.748 [0.728, 0.768]
Boosting	**0.533** [0.517, 0.550]	**0.757** [0.738, 0.775]

Table 6: Test metrics with 95% intervals (1,000 bootstrap samples of test expeditions).

Training set: 43,916 climbers from 7,906 expeditions. Log-loss advantage of boosting: 0.010 over the random forest [0.003, 0.016], 0.046 over the tree [0.035, 0.056], 0.049 over the elastic net [0.039, 0.060]. No interval contains zero.

Backup: confusion matrices of all models

Model	Cut-off	TP	FN	FP	TN	Sensitivity	Specificity
Elastic net	0.50	546	3,035	336	7,216	0.15	0.96
Pruned tree	0.50	1,223	2,358	803	6,749	0.34	0.89
Random forest	0.50	1,359	2,222	713	6,839	0.38	0.91
Boosting	0.50	1,531	2,050	863	6,689	0.43	0.89
Boosting	0.34	2,457	1,124	2,347	5,205	0.69	0.69

Table 7: Test-set confusion matrices; positive class: summit reached.

3,581 of 11,133 test climbers reached the summit. A cut-off of 0.5 favours specificity because only a third of the climbers summit. The cut-off can be set by the cost of each error; log loss and Brier score do not depend on it.

Backup: calibration of all four models

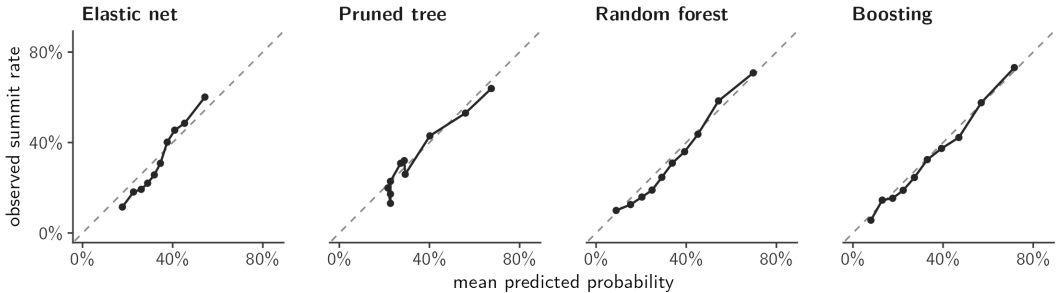


Figure 13: Observed vs. predicted summit rate by decile (test set). Dashed: perfect calibration.

All models lie close to the diagonal. The elastic net and the tree spread their predictions less (20% to 60%); random forest and boosting range from below 10% to above 70%.

Backup: elastic-net coefficient paths

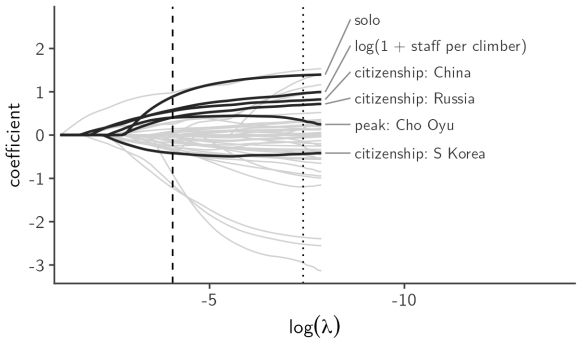


Figure 14: Coefficient paths. Dashed: $\lambda_{\text{1-SE}}$; dotted: $\lambda_{\min}$.

- CV deviance is 1.193 for every $\alpha > 0$ and 1.195 for ridge, so the choice of α hardly matters
- The tree reaches its minimum CV error at 64 leaves; the 1-SE rule gives 30

Backup: tuning grids

m	Node size	CV log loss	OOB
7	**100**	**0.567**	0.499
4	100	0.568	0.508
4	25	0.569	0.475
11	100	0.569	0.497
7	25	0.572	0.467
11	25	0.576	0.467
2	1000	0.590	0.578

Table 8: Random forest (selected rows of the 4×4 grid).

Depth J	Shrinkage ν	Trees M	CV log loss
6	**0.02**	**2,850**	**0.562**
4	0.02	2,950	0.564
6	0.1	350	0.565
4	0.1	1,000	0.567
2	0.1	1,100	0.572
2	0.02	3,000	0.572

Table 9: Gradient boosting, best number of trees per setting.

Both tuned by grouped 5-fold CV. OOB would select the smallest node size.

Backup: variable importance

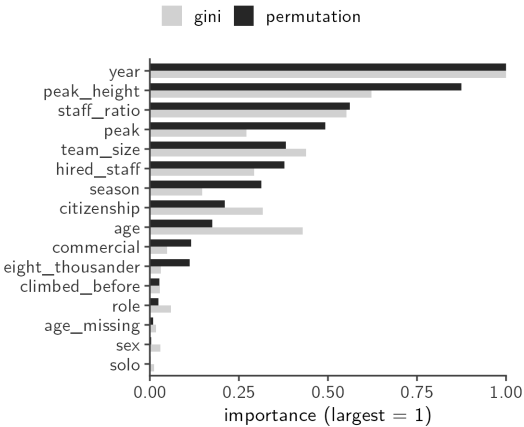


Figure 15: Random forest. Gini overrates age (0.43 vs. 0.17).

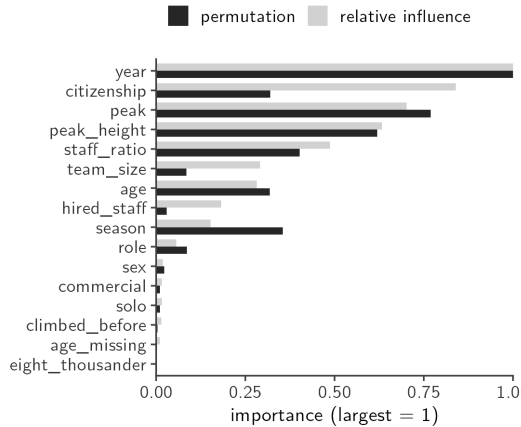


Figure 16: Boosting. Relative influence overrates citizenship.

Backup: one climber, different plans

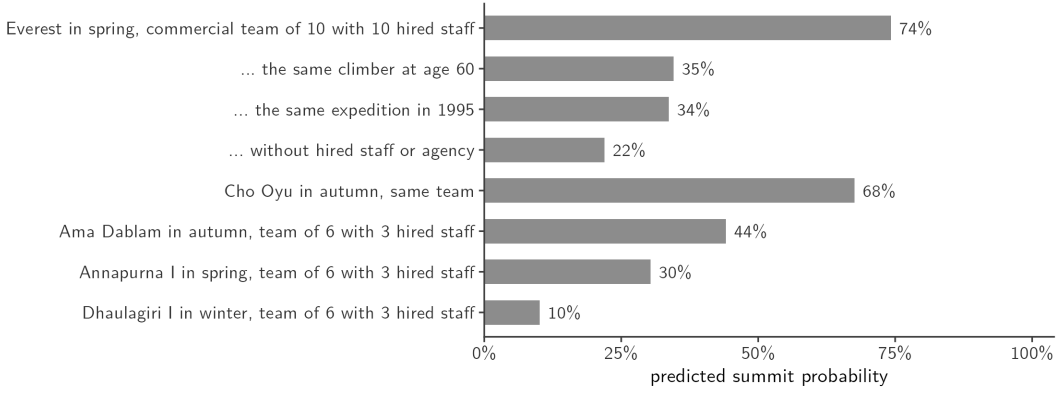


Figure 17: Boosting predictions for a 40-year-old US climber in 2019 under different plans.

These are model predictions, not causal effects: climbers who go without hired staff also differ in other respects.

Backup: time split, all models

Model	Log loss	AUC	Mean predicted
Summit rate only	0.740	0.500	32%
Elastic net	0.703	0.627	34%
Pruned tree	0.668	0.648	42%
Random forest	0.692	0.638	34%
Boosting	0.713	0.661	32%
Observed			47%

Table 10: Trained on 1950–2014, evaluated on 2015–2019 (7,258 climbers).

Boosting ranks future climbers best (highest AUC) but has a worse log loss than the tree, because all models underestimate the higher summit rates after 2015. The main test set measures performance on new expeditions from the same period.

Backup: random vs. grouped cross-validation

Model	Random 5-fold CV		Grouped 5-fold CV		Test set	
	Log loss	AUC	Log loss	AUC	Log loss	AUC
Random forest	0.508	0.807	0.568	0.727	0.543	0.748
Boosting	0.519	0.789	0.562	0.735	0.533	0.757

Table 11: Same tuned settings, folds formed at random (climbers) or by expedition.

- Random folds put team-mates on both sides and make the models look 0.04–0.06 better in log loss
- The test set is one 20% draw of expeditions and somewhat easier: summit rate 32.2% vs. 34.2%, baseline log loss 0.629 vs. 0.643
- Relative to its baseline, boosting improves the log loss by 12.5% in grouped CV and by 15% on the test set

Backup: does the penalty help?

Model (same 49 terms)	Non-zero terms	Test log loss	Test AUC
Logistic regression, no penalty	49	0.575	0.701
Elastic net, $\lambda_{\min}$	48	0.575	0.701
Elastic net, $\lambda_{\text{1-SE}}$ (used)	38	0.582	0.695

Table 12: Linear models on the test set.

- With 55,049 climbers and 49 terms the penalty adds little: $\lambda_{\min}$ equals the unpenalised fit
- The 1-SE rule gives up 0.007 in log loss for 11 fewer terms
- Random forest and boosting are 0.03–0.04 better than every linear version, so the conclusions do not change

Backup: is hired staff only a proxy for era or commercial expeditions?

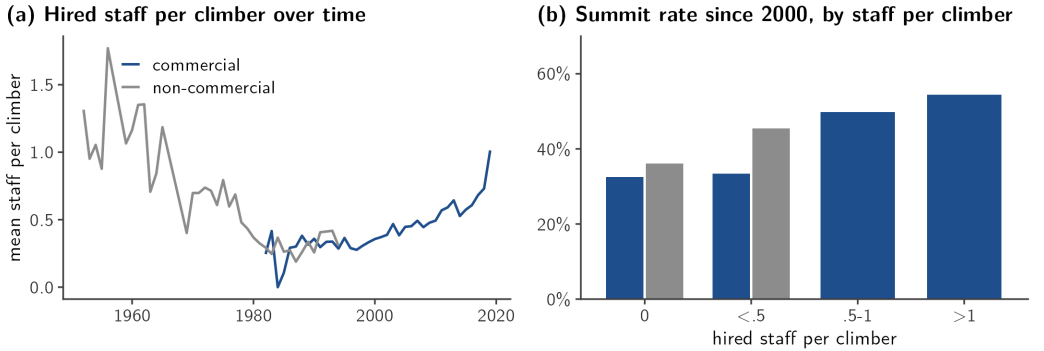


Figure 18: Staff per climber over time and summit rate by staff per climber since 2000. Groups with fewer than 50 climbers omitted.

- 82% of climbers are on commercial expeditions; staff per climber has risen since the late 1990s
- Within commercial expeditions since 2000, the summit rate still rises with staff: 32% (none) to 54% (more than one per climber)
- Staff carries information beyond era and commercial status, but this remains an association

Backup: why no neural network?

- The brief asks for at least three course methods; we use four, from linear to boosting
- The data are tabular with 16 mostly categorical predictors, where tree ensembles are usually the strongest learners
- Boosting with $J = 6$ already captures non-linear effects and interactions
- A neural network needs input scaling, dummy encoding and many more tuning choices (hidden units, weight decay, epochs), and is harder to interpret
- Possible extension: a single-hidden-layer network (`nnet`) tuned by the same grouped CV

Backup: variable definitions

Variable	Definition
Success (target)	Climber reached the main summit
Year, season	Year and season of the expedition (spring, summer, autumn, winter)
Peak, height	Peak (10 most climbed + other) and its height in metres; 8000 m peak indicator
Climbed before	Peak had a recorded first ascent before the expedition (93% of climbers)
Team size	Non-hired members of the expedition
Hired staff	Hired staff above base camp; staff per climber = hired staff / team size
Commercial	Expedition organised through a trekking agency (82% of climbers)
Solo	Climber listed as solo (117 climbers)
Age, sex, citizenship, leader	Age (2.6% missing), sex, citizenship (15 most common + other), leader role

Table 13: The 16 predictors. All are known before departure.

Backup: data problems and their handling

Problem	Handling
Expedition id KANG10101 used twice (1910, 2010)	Join expeditions on id and year
First ascent year “201”	Set to missing
Age missing (1,417 climbers, 2.6%)	Training median + missing indicator
213 citizenships, 381 peaks	Top 15 / top 10 by training frequency, rest “other”
525 role names	Keep climbing roles; climber vs. leader
Team size and hired staff strongly skewed	$\log(1+x)$ in the linear model; no transformation for trees
Members who never attempted the climb	Removed from the sample

Table 14: Data quality issues found in the exploratory analysis.